E3.18
AO12

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2010

MSc and EEE PART III/IV: MEng, BEng.and ACGI

MICROWAVE TECHNOLOGY

Wednesday, 5 May 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Lucyszyn
Second Marker(s) : C. Papavassiliou

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Special instructions to candidates

Permeability of free space, $\mu_o = 4\pi \times 10^{-7}\ H/m$

Permittivity of free space, $\varepsilon_o \approx 8.854\ pF/m$

The Questions

1. There is an analogy between the characteristics of electromagnetic waves as they propagate within a homogenous material (for an unbounded plane wave as it travels in one-dimensional space) and down a transmission line (using the telegrapher's equations).

   a) For a uniform electromagnetic plane wave travelling at normal incidence to the surface of a semi-infinite half space, derive the general form of the intrinsic impedance η_1 for a homogeneous material and the corresponding propagation constant γ, for complex constitutive terms.

   [Hint: the constitutive terms are the relative magnetic permeability $\mu_r = \mu / \mu_o \equiv \mu_r' - j\mu_r''$ *; relative electric permittivity due to molecular and atomic resonances,* $\varepsilon_r = \varepsilon / \varepsilon_o \equiv \varepsilon_r' - j\varepsilon_r''$ *; and intrinsic bulk conductivity due to free charge carriers,* $\sigma \equiv \sigma' - j\sigma''$ *].*

   [2]

   b) Give the general form of the characteristic impedance *Zo* and propagation constant γ for a uniform transmission line with infinite length.

   [2]

Electromagnetic wave propagation within a homogeneous material can be represented by the textbook equivalent one-dimensional uniform transmission line (or distributed-element) model shown in Figure 1, terminated by the secondary line parameter characteristic impedance *Zo*.

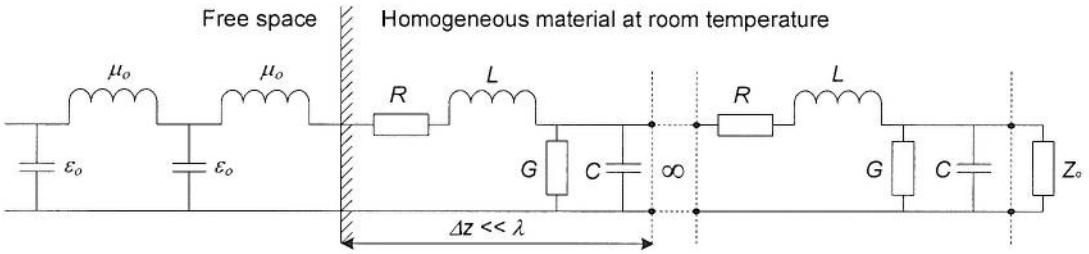


Figure 1. Generic equivalent transmission line model for a homogeneous material

   c) Using the expressions from 1(a) and 1(b), determine equivalent primary line (i.e. distributed-element) parameters for this generic case. For the elementary lumped-element circuits to be valid, what is the relationship between the distance of propagation Δz and the wavelength λ of the electromagnetic wave within the homogeneous material?

   [2]

The intrinsic conductivity of a metal can be expressed by the following equation:

$$\sigma_R = \frac{\sigma_o}{1 + j\omega\tau}$$

where σ_o is the bulk conductivity at dc, the angular frequency $\omega = 2\pi f$, where f is the frequency of the driving electromagnetic fields, and the phenomenological scattering relaxation time for the free electrons is τ.

d) With the over-simplified classical skin-effect model, familiar to microwave engineers, electron scattering relaxation and displacement current terms are ignored (i.e. $\omega\tau \cong 0$ and $\omega\varepsilon_o \cong 0$, respectively).

For normal metals at room temperature, determine the textbook expressions for the:

(i) constitutive parameters [1]

(ii) characteristic impedance [2]

(iii) propagation constant [1]

(iv) distributed-element parameters [1]

Draw the equivalent one-dimensional uniform transmission line model. [2]

What, if any, are the relationships between the distributed-element parameters and the operating temperature and frequency? [1]

e) For gold at room temperature, $\sigma_o = 4.517 \times 10^7$ *[S/m]*. Calculate the following at a frequency of 1 THz:

(i) characteristic impedance [1]

(ii) surface inductance [1]

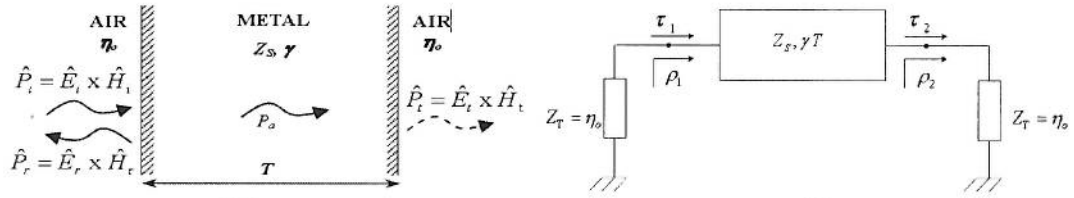
(iii) propagation constant [1]

(iv) wavelength [1]

(v) maximum recommended value of Δz [1]

(vi) elementary lumped-element circuit values using Δz given in (v) [1]

2. Consider an infinite single planar shield in air, having a physical thickness T, with uniform plane wave at normal incidence, as illustrated in Figure 2.



(a) (b)

Figure 2. Uniform plane wave at normal incidence to an infinite single planar shield in air: (a) physical representation; and (b) equivalent 2-port network model (all variables have their usual meaning)

a) Give applications for metal shielding at radio frequencies. Explain why shielding should be made as thin as possible.

[2]

b) Show expressions for the dielectric→metal and metal→dielectric boundary voltage reflection coefficients ρ and transmission coefficients τ.

[Hint: At each boundary, the voltage transmission coefficient is equal to the voltage reflection coefficient plus unity].

Express the corresponding impedances in terms of the constitutive terms μ, ε and σ.

[3]

c) For the simple shielding scenario in Figure 2, the transient response for the overall forward voltage-wave transmission coefficient and overall input voltage-wave reflection coefficient can be derived by summing all possible paths of propagation between the input and output ports.

Given the general mathematical relationship $\sum_{i=0}^{\infty}(x)^i = 1/(1-x)$ where $i \in [0, 1, 2..\infty]$, prove that the steady-state overall forward voltage-wave transmission coefficient and overall input voltage-wave reflection coefficient are respectively given by the following:

$$\tau_1 \cdot \left[\frac{e^{-\gamma T}}{1-\left(e^{-\gamma T}\rho_1\right)^2}\right] \cdot \tau_2 \qquad (2.1) \qquad \text{and} \qquad \rho_1 \cdot \left[\frac{1-e^{-2\gamma T}}{1-\left(e^{-\gamma T}\rho_1\right)^2}\right] \qquad (2.2)$$

[5]

d) The transmission power isolation that results from a shield can be defined by its shielding effectiveness (SE), which is equal to the magnitude of equation (2.1) squared. Shielding effectiveness in decibels, SE_{dB}, can be broken down into three separate terms; each representing the following phenomena of cross-boundary reflections R_{dB}, absorption A_{dB} and multiple reflections, M_{dB}:

$$SE_{dB} \equiv R_{dB} + A_{dB} + M_{dB} \qquad (2.3)$$

Briefly explain each term and, using equation (2.1), give the exact expression for each. Then, simplify each term, quoting any assumptions.

[6]

e) For gold at room temperature, $\sigma_o = 4.517 \times 10^7$ *[S/m]*. Calculate the three terms in 2(d) and overall shielding effectiveness for a shield having a thickness equal to the wavelength in the metal and a frequency of 1 THz.

[4]

3. Consider a balanced transmission line feeding a simple dipole antenna.

a) If the dielectric material between, and completely surrounding, the current carrying metal conductors is non-magnetic and has a dielectric constant of 4.6, calculate the phase velocity of a TEM-wave as it propagates down the transmission line, inside the dielectric.

[4]

b) Valence electrons within the current-carrying metal conductors are free to travel. What is the name commonly given to the mean velocity at which these electrons travel and give the order of magnitude for its value in high conductivity metals?

[2]

c) Within the current-carrying metal conductors an evanescent wave propagates. Calculate the phase velocity of this wave if the metal is made from gold (having a bulk DC conductivity $\sigma_o = 4.517 \times 10^7$ *[S/m]*) at 2 GHz.

[4]

d) Within the metal conductors, the current density is given by, $Jc = N\, e\, v_d$, where electron charge is $e = 1.6 \times 10^{-19}$ *C*. For gold at room temperature, the number of valence electrons is $N = 5.90 \times 10^{22}$ */cm*^{3}. If the metal conductor on a monolithic integrated circuit has a rectangular cross-sectional of 10 x 2 μm^2 and the current flowing is 50 mA, calculate the time average drift velocity, v_d.

[4]

e) For a half-wave dipole in free-space, calculate the distance to the boundary of the radiating far field at 2 GHz. In relation to the speed of light, comment on the phase velocities of the electric field within both the near-field and radiating far field.

[6]

4.

a) A microwave oven operates at 1.9 GHz. If the internal dimensions of the oven are 40 cm x 40 cm x 30 cm, illustrate the E-field energy distribution inside the oven when it is empty. What happens to this distribution when food is placed inside the oven and what is the role of the turntable?

[5]

b) With a peak E-field intensity of 300 V/m, estimate the average power dissipated per unit volume in a portion of meat having a dielectric constant of 40 and loss tangent of 0.5.

[5]

c) If the portion of meat in 4(b) has a mass of 0.75 kg and dimensions of 15 cm x 15 cm x 15 cm and is assumed to have a specific heat of 6 kJ/kg K, estimate the approximate cooking time to raise the temperature of the meat from 0 °C to 90 °C.

[5]

d) Given the complex wavenumber, derive a suitable equation for skin depth from first principles, and calculate this for the meat characterised in 4(b). If the portion of meat is thicker than a few skin depths, what action can be taken to ensure that the meat is cooked more uniformly?

[5]

5.

a) Draw the top and cross-sectional views of a conventional depletion-mode MESFET. Indicate the physical dimensions of gate length Lg, gate width Wg, and depth of depletion d within the doped channel.

[4]

b) Give the simplest expression for the gate transit time τ_g, in terms of the carrier saturation drift velocity v_{sat} and gate length. Show how the gate transit time relates to the cut-off frequency f_T and explain the corresponding difference between GaAs and silicon performances.

[5]

c) Give the simplest expression for the cut-off frequency f_T, in terms of the intrinsic transconductance g_{mo} and gate-source capacitance C_{gs}.

[1]

d) Given a GaAs MESFET having two gate fingers, with each $Lg = 0.5\ \mu m$, $Wg = 60\ \mu m$ and $d = 0.1\ \mu m$, if the measured intrinsic transconductance is 15 mS and the extrapolated cut-off frequency is 21 GHz, calculated the following to a first-order of approximation:

(i) carrier saturation drift velocity

[1]

(ii) gate transit time

[1]

(iii) gate-source capacitance

[1]

(iv) effective dielectric constant of the doped channel

[5]

What is the dielectric constant of undoped GaAs and comment on why the value calculated in (iv) is different.

[2]

6. Figure 6 shows a microphotograph of a 20 to 40 GHz MMIC amplifier.

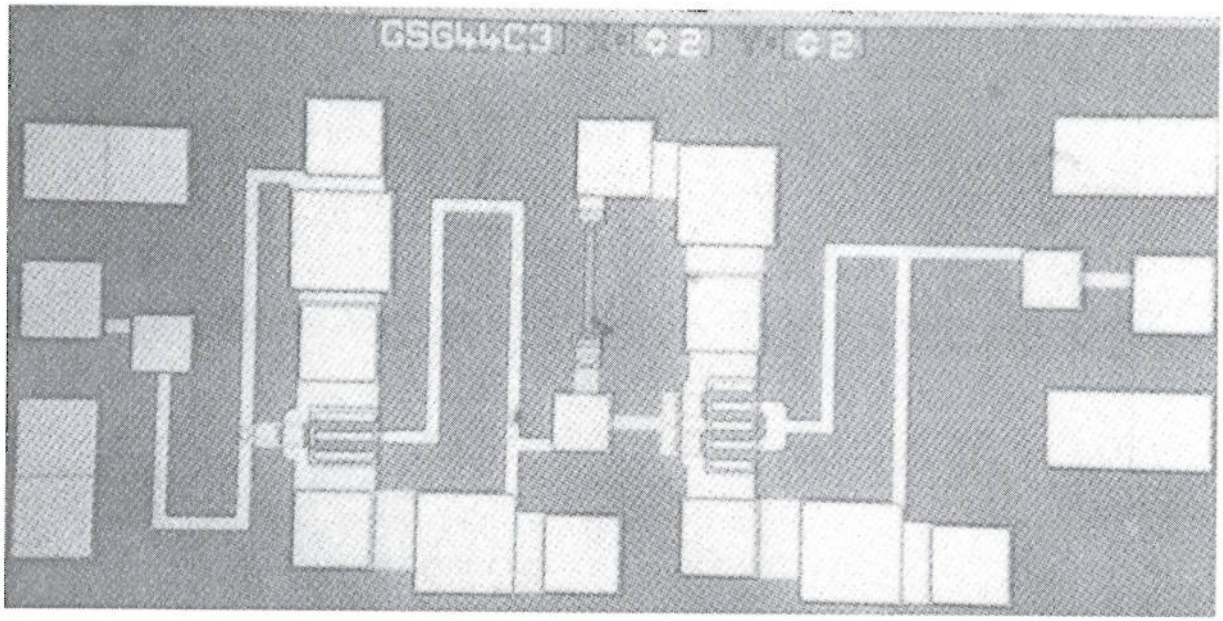


Figure 6. Microphotograph of a 20 to 40 GHz MMIC amplifier

a) By inspection, and showing as much detail as possible, draw the equivalent circuit model for the MMIC shown in Figure 6. Justify all assumptions.

[5]

b) From the circuit extracted from 6(a), answer the following :

(i) What active device technology is being employed? Estimate the periphery of each active device. *[Hint: As a reference, the probe and bond pads are 120 x 120 μm^2 in size].*

[1]

(ii) Explain how biasing of the active devices is being performed.

[1]

(iii) Explain how both DC and RF grounding are being implemented.

[1]

(iv) Explain why spiral inductors are not used.

[1]

c) Briefly describe the following three advanced foundry technologies and give examples of what they are used for:

(i) Multilayer

[2]

(ii) Micromachining

[2]

(iii) RF MEMS

[2]

d) Briefly compare and contrast MMIC and HMIC technologies in terms of cost, reproducibility, reliability, mass, bandwidth, density, active technologies, turn-around time, resources.

[5]

The Solutions for E3.18, 2010

Model answer to Q 1(a): New Derivation

A 012

$$\eta_I = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \sqrt{\frac{(\omega\mu_o\mu_r'') + j\omega(\mu_o\mu_r')}{(\sigma' + \omega\varepsilon_o\varepsilon_r'') + j\omega\left(\frac{-\sigma''}{\omega} + \varepsilon_o\varepsilon_r'\right)}} \quad [\Omega]$$

$$\gamma \equiv \frac{j\omega\mu}{\eta_I} = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} = \sqrt{\left[(\omega\mu_o\mu_r'') + j\omega(\mu_o\mu_r')\right]\left[(\sigma' + \omega\varepsilon_o\varepsilon_r'') + j\omega\left(\frac{-\sigma''}{\omega} + \varepsilon_o\varepsilon_r'\right)\right]} \quad [m^{-1}]$$

[2]

Model answer to Q 1(b): Bookwork

$$Zo = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad [\Omega]$$

$$\gamma \equiv \frac{R + j\omega L}{Zo} = \sqrt{(R + j\omega L)(G + j\omega C)} \quad [m^{-1}]$$

[2]

Model answer to Q 1(c): New Derivation

The equivalent primary line (i.e. distributed-element) parameters for this generic case are given by the following:

$$R \equiv \omega\mu_o\mu_r'' \quad [\Omega/m]; \; L \equiv \mu_o\mu_r' \quad [H/m]; \; G \equiv \sigma' + \omega\varepsilon_o\varepsilon_r'' \quad [S/m]; \; C \equiv \frac{-\sigma''}{\omega} + \varepsilon_o\varepsilon_r' \quad [F/m]$$

For the elementary lumped-element circuits to be valid, the distance of propagation Δz must be very much shorter than the wavelength λ of the electromagnetic wave within the homogeneous material.

[2]

Model answer to Q 1(d): New Derivation

(i) The constitutive parameters are represented by the following: $\mu \to \mu_o$; $\varepsilon \to \varepsilon_o \cong 0$; $\sigma \to \sigma_o$

[1]

(ii)

$$Zo_o = \sqrt{\frac{R_o + j\omega L_o}{G_o + j\omega C_o}}$$

$$Zo_o \Rightarrow \eta_{Io} = \sqrt{\frac{j\omega\mu_o}{\sigma_o}} \Rightarrow Z_{So} \equiv (R_{So} + jX_{So})\,[\Omega/square]$$

where $X_{So} \equiv R_{So}$; $R_{So} \equiv \Re\{Z_{So}\} = \sqrt{\frac{\omega\mu_o}{2\sigma_o}}\,[\Omega/square]$; $L_{So} \equiv \frac{\Im\{Z_{So}\}}{\omega} = \sqrt{\frac{\mu_o}{2\omega\sigma_o}}\,[H/square]$

[2]

(iii)

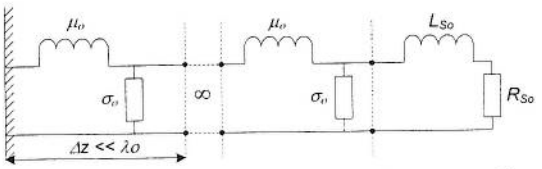
$$\gamma_o \equiv \frac{R_o + j\omega L_o}{Zo_o} = \sqrt{(R_o + j\omega L_o)(G_o + j\omega C_o)}$$

[1]

(iv) Therefore, the distributed-element parameters for the classical skin-effect model become:

$$R_o = 0 \;\; ; \;\; L_o = \mu_o \;\; ; \;\; G_o = \sigma_o \;\; ; \;\; C_o = \varepsilon_o \cong 0$$

[1]



Equivalent transmission line model for the classical skin-effect model

[2]

It is interesting to note that these parameters are frequency invariant, but with $\sigma_o \propto \tau$ there is a strong temperature dependence.

[1]

Model answer to Q 1(e): Calculated Example

For gold at room temperature, $\sigma_o = 4.517*10^7$ [S/m] and $\mu_o = 4\pi \times 10^{-7}\ H/m$. Calculate the:

(i) characteristic impedance Zo = 0.418 (1 + j) [Ω]

[1]

(ii) surface inductance $Ls = \Im\{Zo\}/\omega = 66.54$ [fH/square]

[1]

(iii) propagation constant $\gamma = \sqrt{\omega\mu_o\sigma_o} = 9.44\ (1 + j)$ [μm^{-1}]

[1]

(iv) wavelength $\lambda = 2\pi / \Im\{\gamma\}$ 0.666 [μm]

[1]

(v) maximum recommended value of Δz = 6.66 nm

[1]

(vi) Elementary lumped-element circuit values using Δz given in (v)

$\mu_o \cdot \Delta z = 8.4\,[fH]$ and $\sigma_o \cdot \Delta z = 300\,[mS]$

[1]

Model answer to Q 2(a): New Derivation

At RF, metal shielding is found in many applications; ranging from the construction of high isolation subsystem partitioning walls, efficient quasi-optical components (e.g. planar mirrors and parabolic reflectors for open resonators and antennas), creating guided-wave structures that have (near-)zero field leakage (e.g. metal-pipe rectangular waveguides and associated closed cavity resonators) and embedding ground planes within compact 3D multi-layer architectures.

Ideally, metal shields should be made as thin as possible, while meeting the minimum values for figures of merit within the intended bandwidth of operation, in order to reduce weight and cost. For reasons of structural integrity, thin metal shielding can be deposited onto either a solid plastic/ceramic or even honeycomb supporting wall. Moreover, thin metal shielding embedded between dielectric layers (e.g. to create conformal ground planes or partition walls) can avoid issues of poor topography when integrating signal lines within 3D multi-layered architectures.

[2]

Model answer to Q 2(b): New Derivation

Within the metal, the propagation constant $\gamma \equiv \alpha + j\beta$, α is the attenuation constant and β is the phase constant. With reference to Fig. 1, at the respective dielectric→metal and metal→dielectric boundaries, the voltage reflection coefficients are:

$$\rho_1 = \frac{Z_S - Z_T}{Z_S + Z_T} \quad \text{and} \quad \rho_2 = \frac{Z_T - Z_S}{Z_S + Z_T} \equiv -\rho_1$$

The corresponding voltage transmission coefficients are given by:

$$\tau_1 \equiv 1 + \rho_1 = \frac{2Z_S}{Z_S + Z_T} \quad \text{and} \quad \tau_2 \equiv 1 + \rho_2 = \frac{2Z_T}{Z_S + Z_T}$$

where $Z_S \equiv R_S + jX_S \rightarrow \eta_1 = \sqrt{\dfrac{j\omega\mu}{\sigma + j\omega\varepsilon}}$ is the surface impedance of the metal, R_S is the surface resistance, X_S is the surface reactance and the termination impedance of this equivalent 2-port network $Z_T \rightarrow \eta_o = \sqrt{\dfrac{\mu_o}{\varepsilon_o}} >> \eta_1$ is the intrinsic impedance of the surrounding air dielectric. The other variables have their usual meaning.

[3]

Model answer to Q 2(c): New Derivation

The overall forward voltage-wave transmission coefficient S_{21} for this simple shielding scenario can be represented by the following transient response solution.

$$S_{21} = \tau_1 \cdot \left[e^{-\gamma T} \cdot \sum_{i=0}^{\infty} \left(e^{-\gamma T} \rho_2 \right)^{2i} \right] \cdot \tau_2 \quad \text{where} \quad i \in [0, 1, 2..\infty]$$

From (3), the steady-state solution is easily shown to be given by the following:

$$S_{21} = \tau_1 \cdot \left[\frac{e^{-\gamma T}}{1 - \left(e^{-\gamma T} \rho_1 \right)^2} \right] \cdot \tau_2 \quad \text{since} \quad \left| e^{-\gamma T} \rho_1 \right| < 1$$

Similarly, the overall input voltage-wave reflection coefficient S_{11} for this simple shielding scenario can be represented by the following transient response solution.

$$S_{11} = \rho_1 + \tau_1 \cdot \left[e^{-2\gamma T} \rho_2 \cdot \sum_{i=0}^{\infty} \left(e^{-\gamma T} \rho_2 \right)^{2i} \right] \cdot \tau_2 \quad \text{where} \quad i \in [0, 1, 2..\infty]$$

From (5), the steady-state solution is easily shown to be given by the following:

$$S_{11} = \rho_1 \cdot \left[\frac{1 - e^{-2\gamma T}}{1 - \left(e^{-\gamma T} \rho_1 \right)^2} \right] \quad \text{since} \quad \left| e^{-\gamma T} \rho_1 \right| < 1$$

[5]

Model answer to Q 2(d): New Derivation

R_{dB} is the combined transmission loss caused by the impedance mismatch reflections at the dielectric→metal and metal→dielectric boundaries:

$$R_{dB} = -20 \log_{10} \left| \tau_1 \cdot \tau_2 \right| = 20 \log_{10} \left| \frac{(Z_{So} + \eta_o)^2}{4 Z_{So} \eta_o} \right| \sim 20 \log_{10} \left| \frac{\eta_o}{4 Z_{So}} \right|$$

[2]

A_{dB} is the absorption (or penetration) loss of the electromagnetic energy as it propagates through the shield in the shortest path:

$$A_{dB} = -20 \log_{10} \left| e^{-\gamma_o T} \right| = 20 \log_{10} \left(e^{T/\delta_{So}} \right) \cong 8.686 \cdot T / \delta_{So}$$

[2]

M_{dB} is a correction factor that takes into account the multiple (theoretically infinite) reflections between metal→dielectric boundaries:

$$M_{dB} = 20 \log_{10} \left| 1 - \left(e^{-\gamma_R T} \rho_{1R} \right)^2 \right| \sim 0$$

[2]

Model answer to Q 2(e): Calculated Example

Surface impedance $Zs = 0.418\,(1 + j)\ [\Omega]$

$$R_{dB} \sim 20 \log_{10} \left| \frac{\eta_o}{4 Z_{So}} \right| = 20 \log_{10} \left| \frac{120\pi}{4 \cdot \sqrt{2} \cdot 0.418} \right| = 44.05\ dB$$

$$T \equiv \lambda = 2\pi \cdot \delta_{So}$$

$$A_{dB} \cong 8.686 \cdot T / \delta_{So} = 8.686 \cdot 2 \cdot \pi = 54.58\ dB$$

$$M_{dB} \sim 0 \qquad \text{when} \quad T > 3\delta_{So}$$

Shielding Effectiveness = 98.63 dB

[4]

Model answer to Q 3(a): Computed Example

With a transmission line supporting TEM-mode propagation, the E- & H-field waves propagate outside the conductors at the phase velocity inside the dielectric, v_{pd}:

$$Vpd = \frac{c}{\sqrt{\mu_r \varepsilon_r}} \rightarrow \frac{3.0x10^8\ m/s}{\sqrt{4.6}} \quad = 1.4x10^8\ m/s$$

where, c = speed of light in a vacuum; μ_r = relative permeability of surrounding dielectric = 1; ε_r = relative permittivity of the surrounding dielectric = 4.6

[4]

Model answer to Q 3(b): Computed Example

The free electrons inside the conductors travel at the Fermi speed, v_f:

$$v_f = \sqrt{\frac{2E_f}{m}} \cong 1.4x10^6\ m/s \quad for\ gold$$

where, E_f = Fermi Energy; m = rest mass of the electron The student does not need to know this detailed calculation, but should appreciate the order of magnitude.

[2]

Model answer to Q 3(c): Computed Example

The E- & H-field waves propagate inside the conductors at the phase velocity, v_{pc}:

$$v_{pc} = \frac{\omega}{\beta} = \omega\delta = \sqrt{\frac{2\omega}{\mu_o \sigma_o}} \cong 2.1x10^4\ m/s \quad for\ gold\ at\ 2\ GHz$$

[4]

Model answer to Q 3(d): Computed Example

Conduction current, $Jc = N\,e\,v_d$, flows when (Ne) electrons drift at a time average drift velocity, v_d:

$$v_d = \frac{1}{N \cdot e}\frac{I}{Area} \cong 0.26\ m/s \quad with\ 50\ mA\ flowing\ in\ the\ track$$

[4]

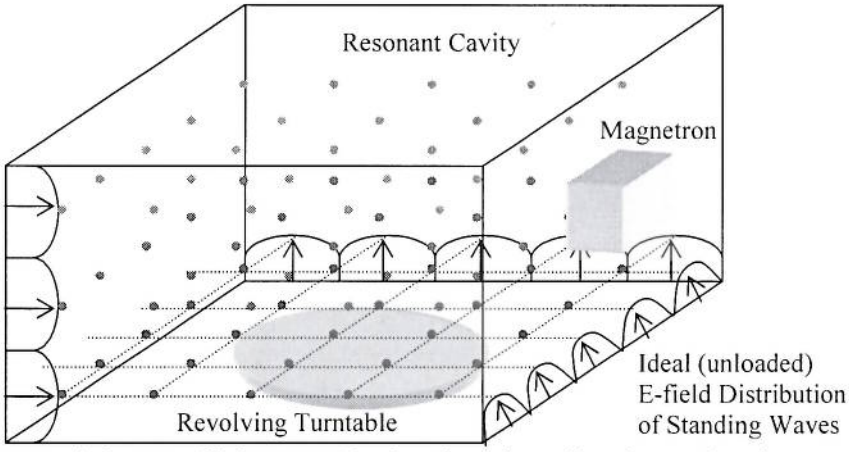
Model answer to Q 3(e): Computed Example

As the radius of the detached electric field loop increases, its curvature straightens out towards a line. With antenna radiation, this spherical wave approximates a plane wave, whereby the E- & H-field and direction of propagation are all mutually orthogonal (i.e. TEM propagation in free space). At a distance greater than $2L^2/\lambda o$ (where L = largest spatial dimension of the antenna's aperture = $\lambda o/2$ in this case), the radiating far field (or Fraunhofer) region exists. What is important is to note that, within this region, the radiation pattern of the antenna is not a function of distance from the antennas. Therefore, the boundary for the radiating far field is $\lambda o/2 = 3 \times 10^8/(2 \times 2 \times 10^9) = 75$ mm. Beyond this boundary, the electric field will be propagating at the speed of light (3×10^8 m/s). However, close to the antenna and well within this boundary the velocity will be greater than the speed of light.

[6]

Model answer to Q 4(a): Bookwork and Computed Example

At 1.9 GHz the wavelength in free space is 157.9 mm. Therefore, there are approximately 5 x 5 x 4 = 100 points of E-field maxima (not shown accurately in the figure).



The E-field distribution illustrated above will be completely altered at all points other than at the cavity walls. Because there are both E-field maxima and minima distributed within the cavity, there will be localised hot-spots and cold-spots in any food. With time, thermal conduction will help to even out these hot and cold spots. In an attempt to avoid hot and sold spots, a turntable below the food or metallic stirring fan above the food "sees" a time and space variant E-field distribution.

[5]

Model answer to Q 4(b): Computed Example

The peak E-field intensity is E_{pk} = 300 V/m. For meat having ε_r' = 40 and tanδ = 0.5.
$\sigma = \omega\varepsilon_0\varepsilon_r'$ tanδ = 2.11 S/m

$\therefore$ Power absorbed per unit volume, $P_v = \sigma E_{RMS}^2|_{dielectric\ interface} \sim 95$ kW/m^3

[5]

Model answer to Q 4(c): Computed Example

Power dissipated (i.e. absorbed) within the food, $P = \dfrac{mS_p\Delta T}{t}\ [W]$

m = mass [g]
S_p = specific heat capacity [kJ/kg K]
ΔT = raise in temperature [K or °C]
t = time [s]

- Power dissipated per unit volume, $P_v = \dfrac{\rho S_p \Delta T}{t}\left[W/m^3\right]$

ρ = mass/volume [g/m^3]

If the meat has dimensions of 15 x 15 x 15 cm^3 then using the result from 4(b):

P_v = 95 kW/m^3 and volume = 3.375 x 10^{-3} m^3, $\therefore P = P_v$ x *volume* = 320.6 W

With m = 0.75 kg, S_p = 6 kJ/kg K, ΔT = 90°C

$$\therefore\ t = \frac{mS_p\Delta T}{P} = \frac{750\times 6\times 90}{320.6} = 1263\ s \cong 21 \text{ minutes}$$

[5]

Model answer to Q 4(d): Computed Example

Propagation constant, $\gamma = \alpha + j\beta \equiv jk$ where $k = \omega\sqrt{\mu\varepsilon}$

$$\therefore\ \gamma = j\omega\sqrt{\mu\left(\varepsilon' - j\varepsilon''\right)} \text{ where } \varepsilon'' = \frac{\sigma}{\omega}$$

$$\gamma = j\omega\sqrt{\mu\varepsilon'}\left(1 - j\frac{\sigma}{\omega\varepsilon'}\right)^{\frac{1}{2}}$$

$$\gamma \approx j\omega\sqrt{\mu\varepsilon'}\left(1 - j\frac{\sigma}{2\omega\varepsilon'}\right)$$

$$\alpha \approx \omega\sqrt{\mu\varepsilon'}\frac{\sigma}{2\omega\varepsilon'} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon'}}$$

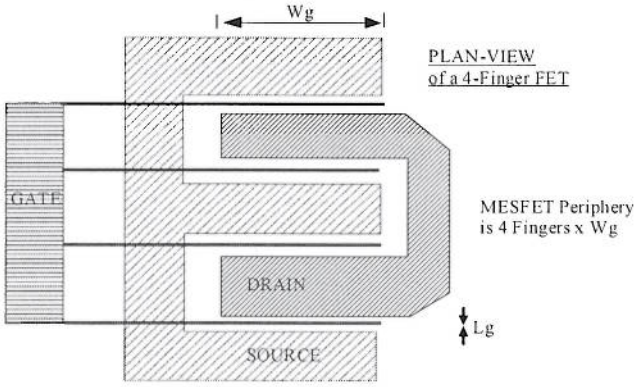
$$\therefore\ \alpha \approx \frac{\sigma}{2}\left(\frac{120\pi}{\sqrt{\varepsilon_r'}}\right)$$

But, skin depth, $\delta = \frac{1}{\alpha}$

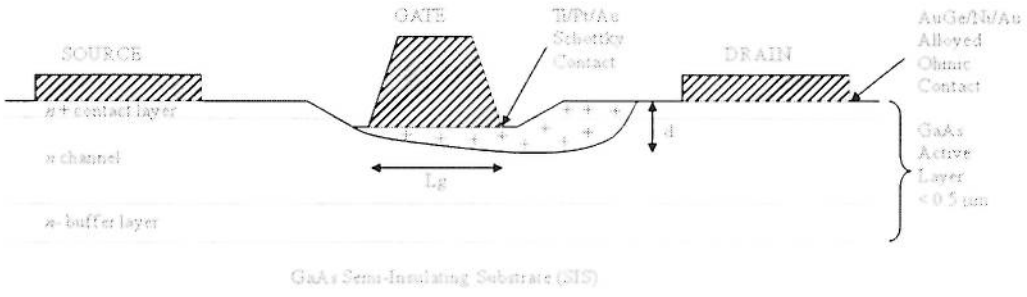
With σ = 2.11 S/m and ε_r' = 40: δ = 1.59 cm. At this depth the power will have fallen to $1/e^2$ or 14% of its surface value. Since the thickness of the meat is approximately 10 skin depths thick, the meat should be cubed and separated so that it can cook more uniformly (e.g. 1.25 inches width/length/height dimensions almost correspond to 2 skin depths).

[5]

Model answer to Q 5(a): Bookwork



Cross-Section of a Recessed-Gate Depletion-Mode
GaAs Metal-Semiconductor Field Effect Transistor (MESFET)



[4]

Model answer to Q 5(b): Textbook Derivation

$$\tau_g = \frac{L_g}{v_{sat}} \qquad f_T = \frac{1}{2\pi\tau_g}$$

$$v_{sat}(GaAs) > v_{sat}(Si)$$
$$\tau_g(GaAs) < \tau_g(Si)$$
$$f_T(GaAs) > f_T(Si)$$

[5]

Model answer to Q 5(c): Textbook Derivation

$$f_T = \frac{g_{mo}}{2\pi Cgs}$$

[1]

Model answer to Q 5(d): Computed Example

(i)

$$v_{sat} = 2\pi f_T Lg = 6.6\times10^6\ cm/s$$

[1]

(ii)

$$\tau_g = \frac{L_g}{v_{sat}} = 7.576\ ps$$

[1]

(iii)

$$Cgs = \frac{g_{mo}}{2\pi f_T} = 113.68\ fF$$

[1]

(iv)

$$Cgs = \varepsilon_o \varepsilon_{r_effective} \frac{Wg\,Lg}{d}$$

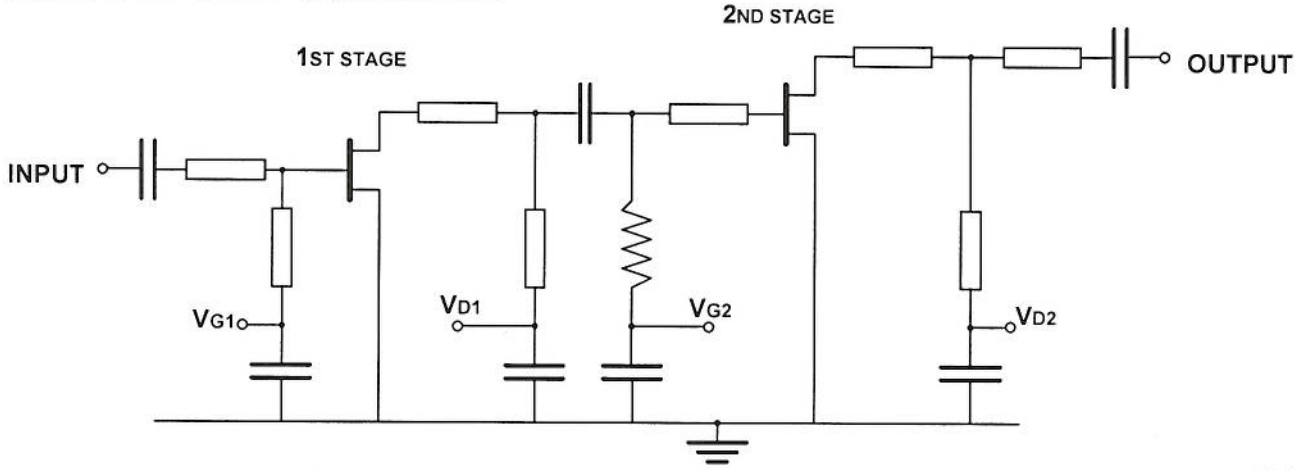
$$\therefore \varepsilon_{r_effective} = \frac{Cgs\ d}{\varepsilon_o\ Wg\ Lg} = 21.4$$

[5]

Undoped GaAs has a dielectric constant of 12.9. The increase of 66% in the effective value calculated in (iv) is due to the doping process of the active layer that creates the channel.

[2]

Model answer to Q 6(a): Bookwork



[5]

Model answer to Q 6(b): Bookwork

(i) The active devices are metal-semiconductor field-effect transistors (MESFETS). The first stage employs a 2-finger 100 micron gate width MESFET, while the second stage is a 4-finger 100 micron gate width MESFET and so is likely to have twice as much power gain.

[1]

(ii) Quarter wavelength microstrip stubs are used on the gate and drain of both transistors, with the exception of the gate to the second stage that employs an isolation resistor. With the former, an RF short circuit that is transformed to an RF open circuit at the transistor.

[1]

(iii) A relatively large shunt capacitor provides the RF short circuits; while through-GaAs vias provide the DC short circuits.

[1]

(iv) Spiral inductors suffer from many unwanted parasitic capacitances, especially at the bends. As a result, the self resonant frequencies will fall within the operating bandwidth of 20 to 40 GHz.

[1]

Model answer to Q 6(c): Bookwork

ADVANCED FOUNDRY TECHNOLOGIES

- Multilayer technology consists of metal and dielectric layers that are deposited on top of the other, without creating air-filled cavities
  - 3-D spiral inductors & transformers, dielectric-filled rectangular waveguides
- Micromachining consists of using one or more semiconductor substrates and using etching techniques to create air-filled cavities
  - membrane suspended transmission lines and coupled-cavity filters, etc
- Micro-electromechanical systems (MEMS) use both multilayer and micromachining technologies to realise moving structures on a chip

RF MEMS is used to implement high performances switches and variable capacitors.

[6]

Model answer to Q 6(d): Bookwork

MMICs	*Hybrid MICs*
Cheap in large quantities, especially economical for complex circuits having many transistors	Simple circuits can be cheaper, automatic assembly is possible
Very good reproducibility	Poor reproducibility due to device placement and bond-wires
Reliable	Hybrids are mostly 'glued' together and so reliability suffers
Small and light	Compact multilayer substrates with embedded passives now available
Less parasitics, more bandwidth & higher frequencies	Poor interconnect can kill the performance of individual components
Space is at a premium, the circuit must be made as small as possible	Substrate is cheap, which allows microstrip to be used abundantly
Very limited choice of component	A vast selection of passive components is available and best transistors can be chosen for LNAs & PAs
Long turn around time (3 months)	Can be very fast (1 week), making multiple iterations possible
Very expensive to start up	Very little capital equipment is required

[5]